Day 108 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Decompose Composite Polygons

Standards: 6.G.A.1

Learning target: I can decompose a polygon into familiar shapes to find area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 23

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01 / Learn

Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure.

Worked example 1

An L-shaped garden is a 4 m by 4 m rectangle with an 2 m by 2 m corner rectangle removed. Sketch and find the garden area.

1. Sketch the figure and label every given length.

2. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle.

3. Outer area 16 minus cutout 4 = 12.

4. State the result in the form requested: 12 m².

Answer: 12 m²

Worked example 2

A 6 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

1. Sketch the figure and label every given length.

2. Outer area = 60.

3. Triangle = 4 × 6/2 = 12.

4. Subtract the removed region exactly once.

Answer: Removed 12 m²; remaining 48 m²

Day 108 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

An L-shaped garden is a 13 m by 11 m rectangle with an 8 m by 6 m corner rectangle removed. Sketch and find the garden area.

02 / Constructed response

An L-shaped garden is a 11 m by 9 m rectangle with an 5 m by 3 m corner rectangle removed. Sketch and find the garden area.

03 / Constructed response

An L-shaped garden is a 14 m by 11 m rectangle with an 8 m by 5 m corner rectangle removed. Sketch and find the garden area.

04 / Constructed response

An L-shaped garden is a 8 m by 9 m rectangle with an 3 m by 4 m corner rectangle removed. Sketch and find the garden area.

05 / Constructed response

An L-shaped garden is a 9 m by 9 m rectangle with an 3 m by 3 m corner rectangle removed. Sketch and find the garden area.

06 / Constructed response

An L-shaped garden is a 14 m by 8 m rectangle with an 8 m by 2 m corner rectangle removed. Sketch and find the garden area.

07 / Constructed response

A 4 m by 7 m rectangle has a right-triangle corner removed with perpendicular legs 2 m and 5 m. Sketch the remaining polygon and find its area.

08 / Two-part numeric entry

A 11 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

09 / Two-part numeric entry

A 8 m by 8 m rectangle has a right-triangle corner removed with perpendicular legs 5 m and 3 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

10 / Constructed response

A 9 m by 6 m rectangle has a right-triangle corner removed with perpendicular legs 3 m and 3 m. Sketch the remaining polygon and find its area.

Day 108 | Unit 08 | Answer key

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1. 95 m²

Outer area 143 minus cutout 48 = 95.

6.G.A.1 | Constructed response

2. 84 m²

Outer area 99 minus cutout 15 = 84.

6.G.A.1 | Constructed response

3. 114 m²

Outer area 154 minus cutout 40 = 114.

6.G.A.1 | Constructed response

4. 60 m²

Outer area 72 minus cutout 12 = 60.

6.G.A.1 | Constructed response

5. 72 m²

Outer area 81 minus cutout 9 = 72.

6.G.A.1 | Constructed response

6. 96 m²

Outer area 112 minus cutout 16 = 96.

6.G.A.1 | Constructed response

7. 23 m²

Whole rectangle 28; remove triangle 5; difference 23.

6.G.A.1 | Constructed response

8. Removed 12 m²; remaining 98 m²

Outer area = 110. Triangle = 4 × 6/2 = 12. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

9. Removed 15/2 m²; remaining 113/2 m²

Outer area = 64. Triangle = 5 × 3/2 = 15/2. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

10. 99/2 m²

Whole rectangle 54; remove triangle 9/2; difference 99/2.

6.G.A.1 | Constructed response